B9339 Systematic Investment Strategies

Homework 3: FX Carry

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Prof. Achilles Venetoulias, Fall 2026 October 15, 2026

Data: hw3.data.carry.FX.dly.xlsx (called hw3.data.FX.dly.xlsx in the assignment), sheet 'Data', 2981 weekday observations from 1999-01-01 to 2010-06-04. Throughout, returns are daily arithmetic USD excess returns on zero-cost portfolios (long side +1, short side −1), rebalanced daily, with no transaction costs.

Summary of answers

(a) The universe is the 10 G-10 currencies including USD (USD carry = 0, return = 0). Each day t the currencies are ranked by carry $c_{i,t} = i_{i,t} - i_{USD,t}$ using either the 1M or the 3M rate; the K = 2 or 3 highest-carry currencies are held long and the K lowest short, with equal, rank-based (K, K−1, …, 1) or proportional-to-carry (distance from the cross-sectional median carry) magnitudes, normalised to +1 long and −1 short. Holdings change through time; typical positions are long NZD and AUD (plus GBP or NOK when K = 3) and short JPY and CHF (plus USD, SEK or EUR when K = 3). Signed weights for all 12 portfolios on 2007-06-29 are in Table 3; full daily histories are exported.

(b) Using a rule fixed in advance (average rank of Sharpe, Calmar and 95% expected shortfall), the best baseline is 3M-K3-EQ: annualised arithmetic mean 0.06413, CAGR 0.06091, volatility 0.09989, Sharpe 0.64198, skewness -0.49551, excess kurtosis 11.33067, maximum drawdown -0.31637 (2007-07-25 to 2009-02-02, not recovered by 2010-06-04). Its 1M twin is a close second (Sharpe 0.64014). The number of currencies matters most: K = 3 beats K = 2 on Sharpe in 6/6 matched pairs and on drawdown in 6/6; the average matched Sharpe range is 0.07702 for K vs 0.04145 for allocation and 0.01736 for tenor. Spreading the book over more currencies diversifies currency-specific crash risk while giving up little carry, which is intuitive.

(c) Carry losses coincide with rising volatility: the same-day correlation of 3M-K3-EQ returns with volatility changes is negative for all six series, strongest for VIX (-0.49472) and JPY3M (-0.37871). Linear regressions of future returns on volatility *levels* are insignificant, but regime splits are not: after low-volatility days the next-day Sharpe is far higher than after high-volatility days. The largest low-minus-high spread is for AUD3M (1.67797 vs 0.20548), followed by AY3M (1.44474 vs 0.27359); this makes AUD3M the candidate. VIX has the strongest contemporaneous link but the weakest regime spread (high-VIX Sharpe 0.70091 exceeds medium 0.20984). The data support a carry / FX-vol / equity-vol link contemporaneously; for conditioning, FX (AUD-related) implied vol is more useful than equity vol. The Jurek reading was not available to me, so I do not attribute specific claims to it.

(d) The rule, identical for all six series, is to be invested on day t+1 only if the latest observed volatility at t is at or below its expanding 75th percentile (computed from prior observations, 252-observation warm-up); otherwise zero exposure. On the common sample 2000-01-04 to 2010-06-04, AY3M-cond is best: Sharpe 1.17806 vs 0.61256 unconditioned, CAGR 0.06978 vs 0.05938, maximum drawdown -0.10212 vs -0.31637, invested 74.70% of days. The (c) candidate AUD3M ranks 3 of 6 (Sharpe 0.95624). So (c) points to the right family of variables (AUD-linked FX implied vol) but not to the exact winner. This is why all six have to be tested: a descriptive full-sample association does not determine the real-time behaviour of a threshold rule, including switch timing, missed rebounds and stale quotes.

(e) I use the 21-day realised volatility of AUD/JPY (X_{AUD}/X_{JPY}) with the same rule. On 2000-01-20 to 2010-06-04 it improves on the baseline (Sharpe 0.90474 vs 0.59469, maximum drawdown -0.13562 vs -0.31637) but trails AY3M-cond (Sharpe 1.14809, drawdown -0.10212). In autumn 2008 the realised-vol signal switched off on 2008-09-09, versus 2008-08-13 for AY3M, and lost -0.05034 over Sep–Dec 2008 versus 0.00000. In August 2007 it reacted earlier. Overall the forward-looking implied measure works better, which matches intuition, though not uniformly.

All results are in-sample over a single period (1999–2010), and picking the best of 12 (or 6) variants is itself an in-sample choice, so none of the rankings should be read as a forecast.

1. Data, conventions and methodology

1.1 Workbook inspection and data dictionary

The workbook has one worksheet ('Data'). The header row is Excel row 1, and every column header was checked programmatically against the column letters given in the assignment (B–J spot, L–T returns plus U = USD ROR, W–AP rates, AR–AW volatilities); all matched. There are 2981 rows. Dates are strictly increasing with no duplicates, and the input was already chronological (it is sorted defensively anyway). Only weekdays appear; consecutive dates are 1 or 3 calendar days apart. The 1-day return cells in L–T are live Excel formulas (e.g. EUR `=B3/B2-1`, JPY `=C2/C3-1`), so they are already expressed as the return of holding the foreign currency.

Table 1: Data dictionary (column-level detail in outputs/tables/data_dictionary.csv).

Excel cols	Headers	Content	Units	Missing
A	Date	Trading date (weekdays, includes non-US holidays)	date	0
B-J	EUR ... SEK	Spot FX vs USD in market convention (see Table 2)	price	0
L-T, U	EUR ROR ... SEK ROR, USD ROR	Workbook 1-day spot return of holding the currency; USD ROR = 0	decimal	1 (first row)
W-AF	EUR 1M ... USD 1M	1-month interest rates, 10 currencies	% p.a.	0
AG-AP	EUR 3M ... USD 3M	3-month interest rates, 10 currencies	% p.a.	0
AR, AS, AT	AY1wk, AY3M, AY6M	AUD/JPY option implied vol, 1-week / 3-month / 6-month	vol pts	0
AU, AV	JPY3M, AUD3M	USD/JPY and AUD/USD 3-month implied vol	vol pts	0
AW	VIX	CBOE VIX (30-day S&P; 500 implied vol)	index pts	108 (US holidays + 1999-01-01)

Table 2: Exchange-rate orientation. Computed returns on USD-per-currency levels reproduce the workbook to machine precision. Merely negating the return of an inverted quote would be wrong by up to 0.00303 on a single day.

currency	workbook quote	X (USD per unit)	max |computed - workbook return|	max error if raw-quote return were negated
EUR	USD per EUR	S	0.00e+00	n/a
JPY	JPY per USD	1 / S	2.22e-16	0.00303
GBP	USD per GBP	S	0.00e+00	n/a
CAD	CAD per USD	1 / S	2.22e-16	0.00160
AUD	USD per AUD	S	0.00e+00	n/a
NZD	USD per NZD	S	0.00e+00	n/a
CHF	CHF per USD	1 / S	2.22e-16	0.00220
NOK	NOK per USD	1 / S	2.22e-16	0.00247
SEK	SEK per USD	1 / S	2.22e-16	0.00248
USD	-	1	0.00e+00	n/a

1.2 Missing, stale and holiday observations

– Spot rates and interest rates have no missing values. Rows exist for every weekday, including holidays such as 1999-01-01 and US holidays; spot quotes on those days are kept as reported. A zero spot change on such a day is a genuine zero observation and is not filled in.

– VIX is blank on 108 dates (US exchange holidays, plus 1999-01-01 before the series starts). For signals I use the *latest observed* VIX (last value carried forward), never before its first observation (1999-01-04). Expanding thresholds are computed only from actually observed values, so carried-forward values are not double-counted. Volatility *changes* in part (c) are set to missing on days with no new observation.

– Stale quotes: AY1wk is unchanged on 99% of days in 1999–2002 and on 79% / 59% of days in 2003 / 2004, so it is effectively a step function early in the sample. AY3M and AY6M are unchanged on roughly 10–40% of days before 2008. I keep the data as supplied but flag this in parts (c) and (d). Interest rates also have runs of unchanged values, which is normal for fixings.

– JPY 3M rates are negative on 376 days and CHF 3M on 1 day; the formulas handle negative rates without modification.

1.3 Returns, carry and accrual

Spot levels are standardised to $X_{i,t}$ = USD per unit of currency i. Inverted quotes are inverted *before* computing returns:

$$X_{i,t} = S_{i,t} \quad (\text{EUR, GBP, AUD, NZD}), \qquad X_{i,t} = 1/S_{i,t} \quad (\text{JPY, CAD, CHF, NOK, SEK}), \qquad X_{USD,t} = 1$$

$$r_{i,\,t+1} = \frac{X_{i,t+1}}{X_{i,t}} - 1, \qquad \Delta_t = \frac{\text{calendar days}(t,\,t+1)}{360}$$

Interest rates are converted from % per annum to decimals. The rate of the chosen tenor observed at t accrues simple interest over the calendar interval to the next observation (ACT/360; Friday to Monday accrues 3 days). The annualised 3M rate is *not* divided by 3: both tenors are annualised quotes, and the tenor only selects which rate curve point is used for ranking and for the daily accrual. Holding a 1M or 3M deposit and rebalancing it daily at that rate is the standard textbook simplification, which I adopt. The funded USD excess return of a 1-USD long position in currency i is

$$e_{i,\,t+1} = (1 + r_{i,\,t+1})(1 + i_{i,\,t}\Delta_t) - (1 + i_{USD,\,t}\Delta_t) \;\approx\; r_{i,\,t+1} + (i_{i,\,t} - i_{USD,\,t})\Delta_t$$

This means: borrow 1 USD at the USD rate, convert to currency i, earn the foreign rate, and convert back at t+1. I use the exact product form, not the first-order approximation, which drops the small cross term $r_{i,t+1} i_{i,t} \Delta_t$. A short position (negative weight) is the mirror trade: borrow currency i at $i_{i,t}$, convert to USD and lend at $i_{USD,t}$. Its P&L is exactly $-e_{i,t+1}$. For USD itself $e_{USD} = 0$. Carry relative to USD and the portfolio return are

$$c_{i,\,t} = i_{i,\,t} - i_{USD,\,t}, \qquad R_{t+1} = \sum_i w_{i,\,t}\, e_{i,\,t+1}, \qquad \sum_{w_{i,\,t}>0} w_{i,\,t} = 1, \quad \sum_{w_{i,\,t}<0} w_{i,\,t} = -1$$

Because the weights sum to zero, R_{t+1} equals the USD return of the long-short book. It is a zero-cost excess return; no interest on posted collateral is added in any comparison. Timing: weights are set from rates known at the close of t and earn the return over (t, t+1]; volatility signals likewise use values known at t. Carry ranking (which currencies have the highest / lowest c) determines exposure direction (long / short); the magnitude rule only distributes the +1 and −1 within each side.

1.4 Treatment of USD

I treat USD as one of the 10 candidate currencies (primary interpretation). The assignment asks for portfolios "based on the USD-based basket of currencies" and the workbook supplies USD 1M and 3M rates in the same blocks as the other nine, plus a USD ROR column of zeros. Both point to a 10-currency G-10 ranking universe in which returns are measured in USD. When USD ranks in the top or bottom K it is simply held, earning zero excess return. Excluding USD (a 9-currency universe) is reported as a sensitivity check in §7; it changes no conclusion.

1.5 Allocation rules

No lecture definitions were supplied with the assignment, so these are my stated conventions, not definitions specified by the professor. Order each side from most to least extreme carry (j = 1 is the highest-carry long or the lowest-carry short), and let m_t be the cross-sectional median carry of the universe:

$$\text{EQ}: \; |w_{(j)}| = \tfrac{1}{K}; \qquad \text{RANK}: \; |w_{(j)}| = \frac{K+1-j}{K(K+1)/2}; \qquad \text{PROP}: \; |w_i| = \frac{|c_{i,\,t} - m_t|}{\sum_{l \,\in\, \text{side}} |c_{l,\,t} - m_t|}$$

RANK gives weights 2/3, 1/3 (K = 2) and 1/2, 1/3, 1/6 (K = 3). PROP measures carry relative to the cross-sectional median rather than to USD. With K ≤ 3 of 10 currencies, every long has carry at or above the median and every short at or below it, so magnitudes are always non-negative, even on days when all selected

currencies have positive (or all negative) carry versus USD. Normalising signed USD carry instead would flip positions whenever, for example, USD is the highest-yielding currency. If every selected currency on a side equals the median (zero denominator), the side falls back to equal weights; this never happens in the sample. Ties in carry are broken by the fixed universe order.

1.6 Performance statistics

$$W_T = \prod_t (1 + R_t), \quad \bar{g} = W_T^{1/N} - 1, \quad \mathrm{CAGR} = (1 + \bar{g})^{260} - 1$$

$$\mu_{ann} = 260\,\bar{R}, \quad \sigma_{ann} = \sqrt{260}\; s_R, \quad SR = \tfrac{\mu_{ann}}{\sigma_{ann}}$$

$$DD_t = \frac{W_t}{\max(1, \max_{s \le t} W_s)} - 1, \quad \mathrm{VaR}_{95} = -q_{0.05}(R), \quad \mathrm{ES}_{95} = -\mathrm{E}[R \mid R \le q_{0.05}(R)]$$

260 observations per year (the sample averages 260.8 weekdays a year). The Sharpe numerator is the annualised arithmetic mean of the daily *excess* return: the USD funding cost is already inside $e_{i,t+1}$, so no further risk-free rate is subtracted. The geometric return comes from compounding the arithmetic daily returns. Annualising the arithmetic mean by 260 (simple) but the geometric mean by compounding means the annualised CAGR can slightly exceed 260×mean when volatility is low. The like-for-like comparison is the daily arithmetic versus daily geometric mean, both reported. VaR and ES are positive numbers meaning losses (fraction of capital per day). Skewness and excess kurtosis are bias-adjusted sample moments. Calmar = CAGR / |maximum drawdown|.

2. Part (a): the 12 carry portfolios

The 12 portfolios are all combinations of tenor (1M, 3M) × K (2, 3 currencies per side) × allocation (EQ, RANK, PROP). They are named `tenor-K-method`, e.g. `3M-K3-EQ`. Each day the steps are: (1) compute $c_{i,t}$ for the 10 currencies from the tenor's rates; (2) select the K highest (long) and K lowest (short); (3) assign magnitudes with the allocation rule; (4) apply the weights to the return over (t, t+1]. Holdings therefore change daily whenever the carry ranking changes, and PROP magnitudes change every day. Table 3 shows one clearly dated example.

Table 3: Signed weights of all 12 portfolios set at the close of 2007-06-29 (applied to the return from 2007-06-29 to 2007-07-02). Long weights sum to +1 and short weights to -1 in every row.

portfolio	long side (signed weight)	short side (signed weight)
1M-K2-EQ	AUD +0.50000, NZD +0.50000	JPY -0.50000, CHF -0.50000
1M-K2-RANK	NZD +0.66667, AUD +0.33333	JPY -0.66667, CHF -0.33333
1M-K2-PROP	NZD +0.67517, AUD +0.32483	JPY -0.66530, CHF -0.33470
1M-K3-EQ	GBP +0.33333, AUD +0.33333, NZD +0.33333	JPY -0.33333, CHF -0.33333, SEK -0.33333
1M-K3-RANK	NZD +0.50000, AUD +0.33333, GBP +0.16667	JPY -0.50000, CHF -0.33333, SEK -0.16667
1M-K3-PROP	NZD +0.54115, AUD +0.26036, GBP +0.19849	JPY -0.57426, CHF -0.28890, SEK -0.13685
3M-K2-EQ	AUD +0.50000, NZD +0.50000	JPY -0.50000, CHF -0.50000
3M-K2-RANK	NZD +0.66667, AUD +0.33333	JPY -0.66667, CHF -0.33333
3M-K2-PROP	NZD +0.67797, AUD +0.32203	JPY -0.66998, CHF -0.33002
3M-K3-EQ	GBP +0.33333, AUD +0.33333, NZD +0.33333	JPY -0.33333, CHF -0.33333, SEK -0.33333
3M-K3-RANK	NZD +0.50000, AUD +0.33333, GBP +0.16667	JPY -0.50000, CHF -0.33333, SEK -0.16667
3M-K3-PROP	NZD +0.54225, AUD +0.25757, GBP +0.20018	JPY -0.57433, CHF -0.28290, SEK -0.14276

Table 4: Inputs on 2007-06-29. NZD, AUD and GBP have the highest carry, and JPY, CHF and SEK the lowest. USD (5.32%) ranks 4th, so it is not selected on this date.

currency	1M rate (%)	3M rate (%)	1M carry vs USD (%)	3M carry vs USD (%)
EUR	4.11500	4.15000	-1.20500	-1.18500
JPY	0.64438	0.68000	-4.67562	-4.65500
GBP	5.92125	5.93500	0.60125	0.60000
CAD	4.47000	4.46880	-0.85000	-0.86620
AUD	6.33000	6.32000	1.01000	0.98500
NZD	8.18500	8.23000	2.86500	2.89500
CHF	2.61500	2.66500	-2.70500	-2.67000
NOK	4.75000	4.71500	-0.57000	-0.62000
SEK	3.66500	3.61950	-1.65500	-1.71550
USD	5.32000	5.33500	0.00000	0.00000

For comparison, on the last signal date (2010-06-03) the 3M-K3-EQ portfolio is long AUD +0.33333, NZD +0.33333, NOK +0.33333 and short JPY -0.33333, CHF -0.33333, SEK -0.33333. In 2010, with US rates near zero, USD enters and leaves the short basket: it is shorted on 53% of 2010 signal dates. This is why holdings must be read as daily histories, not as one fixed list.

Table 5: Selection frequency (% of signal dates on which a currency is in the long or short set; it is the same for all three allocation methods). JPY and CHF are almost always funding currencies and NZD/AUD almost always investment currencies. USD is an investment (long) currency in 1999-2000 and 2006, when US rates were high, and a funding (short) currency in 2001-2004 and from late 2008.

currency	long 1M K=2	long 1M K=3	long 3M K=2	long 3M K=3	short 1M K=2	short 1M K=3	short 3M K=2	short 3M K=3
EUR	0.00	0.00	0.00	0.00	0.00	13.08	0.00	11.71
JPY	0.00	0.00	0.00	0.00	99.03	100	98.96	100
GBP	6.71	47.33	6.94	49.48	0.00	0.00	0.00	0.00
CAD	0.00	2.11	0.00	1.44	0.10	1.58	2.08	6.37
AUD	59.11	77.32	60.35	75.44	0.00	0.00	0.00	0.00
NZD	86.25	89.84	84.80	89.57	0.00	0.00	0.00	0.00
CHF	0.00	0.00	0.00	0.00	99.06	100	94.80	99.36
NOK	37.07	60.01	35.66	59.28	0.00	5.33	0.00	5.94
SEK	0.00	0.13	0.00	0.00	0.00	30.02	2.15	36.67
USD	10.87	23.25	12.24	24.79	1.81	49.98	2.01	39.95

Table 6: How often the selected currencies change. 3M rates are noisier, so the 3M ranking changes about three times as often.

config	days with composition change	% of days	avg days between changes	distinct long sets	distinct short sets
1M K=2	59	1.98	50.51	6	4
1M K=3	66	2.21	45.15	10	5
3M K=2	199	6.68	14.97	6	6
3M K=3	196	6.58	15.20	8	7

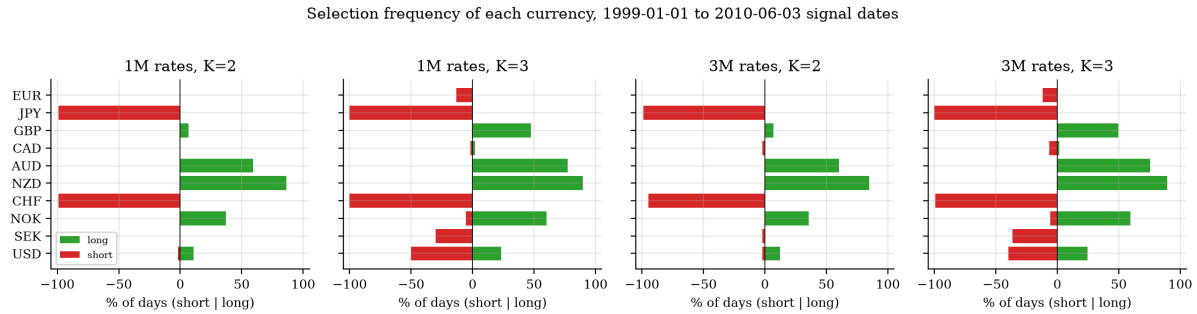


Figure 1: Selection frequency by currency (green = long, red = short).



Figure 2: Daily signed weights for the three 3M-K3 allocation variants. EQ and RANK change only when the ranking changes; PROP moves continuously with the carry spreads.

The file outputs/histories/baseline_weights_daily.csv holds every signed weight (columns `strategy|currency`, indexed by signal date) and baseline_returns_daily.csv holds the 12 return streams (indexed by return date). Every day and every portfolio was validated: exactly K longs and K shorts, long weights sum to +1 and short weights to −1, every long has carry at or above every unselected currency, and every short at or below (outputs/tables/weight_validation.csv).

3. Part (b): comparing the return distributions

All 12 portfolios are evaluated on the same 2980 daily returns (1999-01-04 to 2010-06-04).

Table 7: Return moments (decimals unless marked %). The highlighted row is the selected best baseline.

portfolio	mean/day (%)	geo mean/day (%)	ann. arith. mean	CAGR	ann. vol	Sharpe	Sortino	skew	excess kurt.	final wealth
1M-K2-EQ	0.02628	0.02337	0.06834	0.06264	0.12289	0.55609	0.75506	-0.56643	7.55484	2.00653
1M-K2-RANK	0.02535	0.02210	0.06590	0.05913	0.12977	0.50781	0.68713	-0.62438	9.18866	1.93171
1M-K2-PROP	0.02715	0.02399	0.07060	0.06434	0.12816	0.55089	0.74691	-0.60147	8.30545	2.04361
1M-K3-EQ	0.02553	0.02346	0.06638	0.06289	0.10370	0.64014	0.87950	-0.44742	9.93000	2.01186
1M-K3-RANK	0.02544	0.02294	0.06614	0.06146	0.11376	0.58139	0.78981	-0.58767	10.06476	1.98109
1M-K3-PROP	0.02635	0.02388	0.06852	0.06404	0.11334	0.60458	0.82035	-0.61664	9.41824	2.03696
3M-K2-EQ	0.02570	0.02259	0.06682	0.06050	0.12693	0.52644	0.71319	-0.57229	11.09562	1.96051
3M-K2-RANK	0.02808	0.02455	0.07300	0.06591	0.13514	0.54014	0.73355	-0.59836	10.83428	2.07833
3M-K2-PROP	0.02736	0.02397	0.07115	0.06430	0.13259	0.53660	0.72622	-0.64194	10.62205	2.04270
3M-K3-EQ	**0.02466**	**0.02274**	**0.06413**	**0.06091**	**0.09989**	**0.64198**	**0.88005**	**-0.49551**	**11.33067**	**1.96923**
3M-K3-RANK	0.02637	0.02383	0.06856	0.06390	0.11487	0.59689	0.81200	-0.57158	11.62087	2.03388
3M-K3-PROP	0.02677	0.02430	0.06960	0.06521	0.11316	0.61510	0.83510	-0.64317	11.04183	2.06284

Table 8: Downside statistics. Every drawdown peaks in July 2007 and troughs on 2009-02-02; none has recovered by the end of the sample. VaR/ES are positive loss fractions per day.

portfolio	max DD	DD peak	DD trough	recovered	VaR95/day	ES95/day	worst day	worst date	% days > 0	Calmar
1M-K2-EQ	-0.38264	2007-07-23	2009-02-02	not recovered	0.01210	0.01998	-0.05445	2008-10-24	55.53691	0.16372
1M-K2-RANK	-0.43032	2007-07-23	2009-02-02	not recovered	0.01222	0.02095	-0.06854	2008-10-24	55.23490	0.13740
1M-K2-PROP	-0.41021	2007-07-23	2009-02-02	not recovered	0.01203	0.02077	-0.06031	2008-10-24	55.70470	0.15685
1M-K3-EQ	-0.31171	2007-07-25	2009-02-02	not recovered	0.00987	0.01651	-0.05181	2008-10-24	55.33557	0.20176
1M-K3-RANK	-0.37220	2007-07-23	2009-02-02	not recovered	0.01089	0.01839	-0.06018	2008-10-24	54.96644	0.16513
1M-K3-PROP	-0.36566	2007-07-23	2009-02-02	not recovered	0.01074	0.01836	-0.05533	2008-10-24	55.53691	0.17514
3M-K2-EQ	-0.42071	2007-07-23	2009-02-02	not recovered	0.01194	0.02054	-0.06523	2008-10-08	56.04027	0.14379
3M-K2-RANK	-0.45106	2007-07-23	2009-02-02	not recovered	0.01246	0.02180	-0.06855	2008-10-24	55.36913	0.14612
3M-K2-PROP	-0.44168	2007-07-23	2009-02-02	not recovered	0.01212	0.02144	-0.06443	2008-10-08	55.87248	0.14558
3M-K3-EQ	**-0.31637**	**2007-07-25**	**2009-02-02**	**not recovered**	**0.00943**	**0.01574**	**-0.05182**	**2008-10-24**	**55.30201**	**0.19251**
3M-K3-RANK	-0.38565	2007-07-23	2009-02-02	not recovered	0.01058	0.01845	-0.06019	2008-10-24	55.73826	0.16569
3M-K3-PROP	-0.37781	2007-07-23	2009-02-02	not recovered	0.01045	0.01818	-0.05764	2008-10-24	55.93960	0.17261

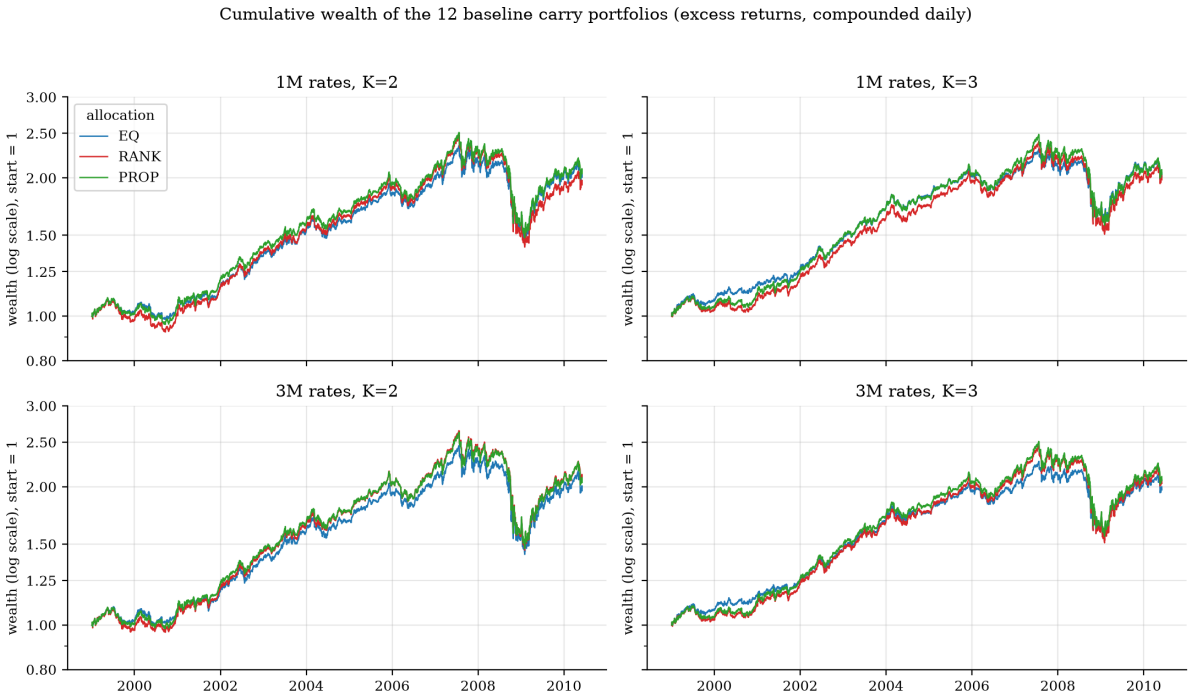


Figure 3: Cumulative wealth (log scale), grouped by tenor and K; colours are allocation methods.

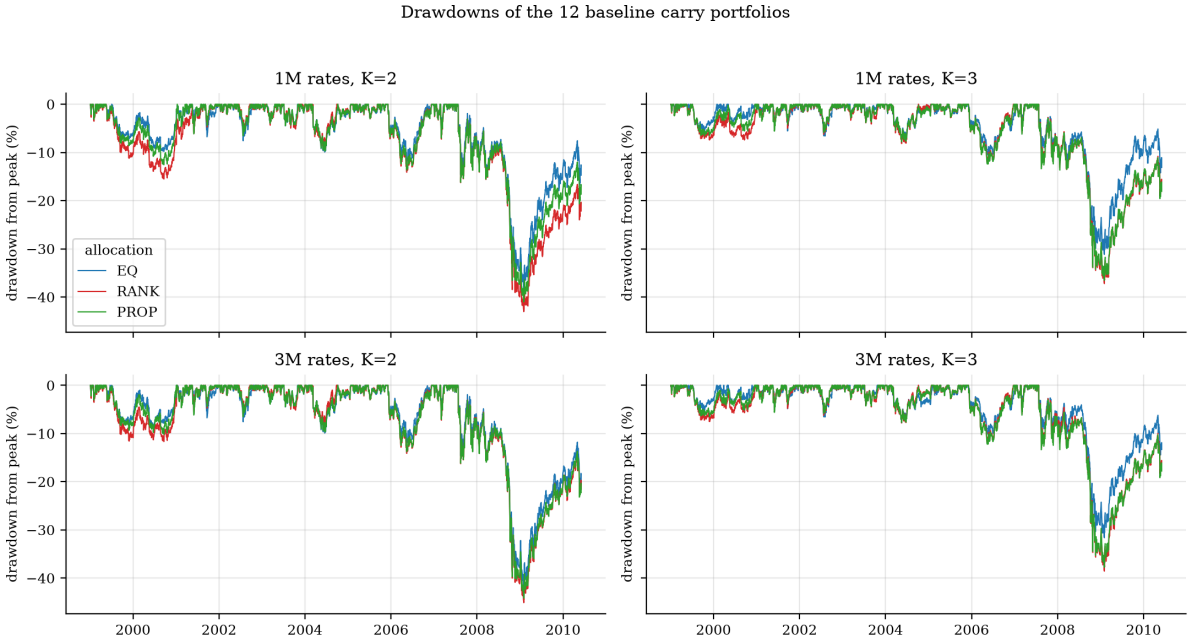


Figure 4: Drawdowns. The 2008 carry crash dominates every portfolio; K = 3 books lose about 31-39% versus 38-45% for K = 2.

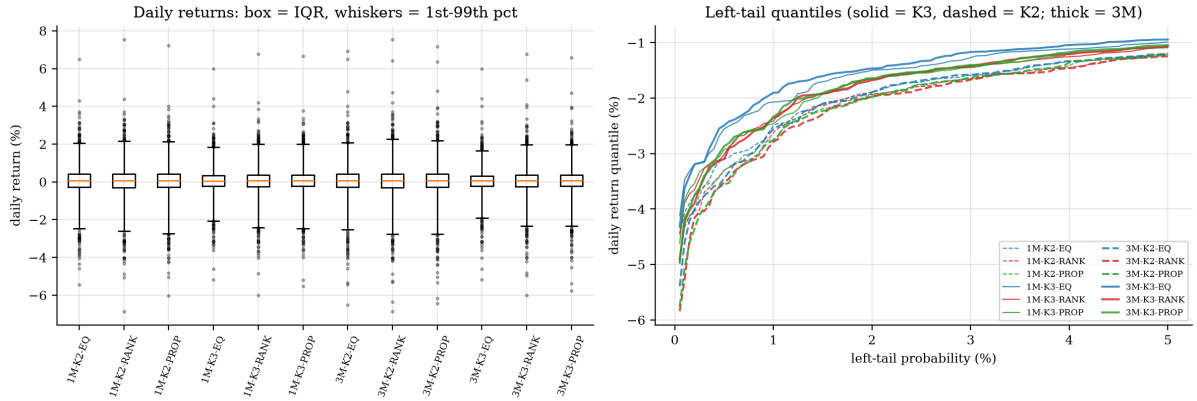


Figure 5: Left: daily return distributions. Right: left-tail quantiles. The K = 3 portfolios (solid) have uniformly smaller tail losses than K = 2 (dashed).

3.1 Matched comparisons

To isolate one factor I compare portfolios that differ *only* in that factor (6 tenor pairs, 6 K pairs, 4 allocation triplets) and average the within-group range (max – min) of each statistic.

Table 9: Average matched range per factor. Larger = the factor matters more. The last column is the lowest correlation between daily returns within a matched group.

factor varied	groups	Sharpe	arith. mean	CAGR	ann. vol	max DD	ES95	min. return corr
rate tenor (1M vs 3M)	6	0.01736	0.00249	0.00243	0.00316	0.02009	0.00051	0.97998
number of currencies (K=2 vs 3)	6	0.07702	0.00216	0.00104	0.01946	0.06787	0.00331	0.96184
allocation (EQ/RANK/PROP)	4	0.04145	0.00468	0.00438	0.01003	0.05195	0.00170	0.97291

Table 10: K = 2 versus K = 3, holding tenor and allocation fixed.

pair	Sharpe K2	Sharpe K3	vol K2	vol K3	CAGR K2	CAGR K3	maxDD K2	maxDD K3	ES95 K2	ES95 K3
1M-EQ	0.55609	0.64014	0.12289	0.10370	0.06264	0.06289	-0.38264	-0.31171	0.01998	0.01651
1M-RANK	0.50781	0.58139	0.12977	0.11376	0.05913	0.06146	-0.43032	-0.37220	0.02095	0.01839
1M-PROP	0.55089	0.60458	0.12816	0.11334	0.06434	0.06404	-0.41021	-0.36566	0.02077	0.01836
3M-EQ	0.52644	0.64198	0.12693	0.09989	0.06050	0.06091	-0.42071	-0.31637	0.02054	0.01574
3M-RANK	0.54014	0.59689	0.13514	0.11487	0.06591	0.06390	-0.45106	-0.38565	0.02180	0.01845
3M-PROP	0.53660	0.61510	0.13259	0.11316	0.06430	0.06521	-0.44168	-0.37781	0.02144	0.01818

Table 11: 1M versus 3M, holding K and allocation fixed.

pair	Sharpe 1M	Sharpe 3M	CAGR 1M	CAGR 3M	maxDD 1M	maxDD 3M
K2-EQ	0.55609	0.52644	0.06264	0.06050	-0.38264	-0.42071
K2-RANK	0.50781	0.54014	0.05913	0.06591	-0.43032	-0.45106
K2-PROP	0.55089	0.53660	0.06434	0.06430	-0.41021	-0.44168
K3-EQ	0.64014	0.64198	0.06289	0.06091	-0.31171	-0.31637
K3-RANK	0.58139	0.59689	0.06146	0.06390	-0.37220	-0.38565
K3-PROP	0.60458	0.61510	0.06404	0.06521	-0.36566	-0.37781

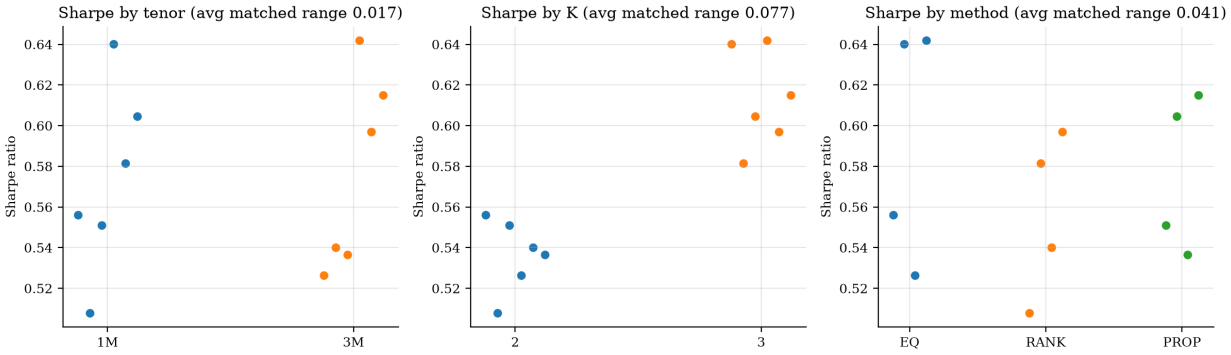


Figure 6: Sharpe ratios grouped by each construction factor.

3.2 Selecting the best baseline

Decision rule, fixed before looking at rankings: average the ranks of (i) the Sharpe ratio (risk-adjusted return), (ii) the Calmar ratio (compounded return per unit of worst drawdown) and (iii) the daily 95% expected shortfall (tail loss), with ties broken by Sharpe. Cumulative return alone is not used.

Table 12: Ranking under the decision rule.

portfolio	Sharpe rank	Calmar rank	ES95 rank	average rank	Sharpe
3M-K3-EQ	**1**	**2**	**1**	**1.33333**	**0.64198**
1M-K3-EQ	2	1	2	1.66667	0.64014
3M-K3-PROP	3	4	3	3.33333	0.61510
1M-K3-PROP	4	3	4	3.66667	0.60458
3M-K3-RANK	5	5	6	5.33333	0.59689
1M-K3-RANK	6	6	5	5.66667	0.58139
1M-K2-EQ	7	7	7	7	0.55609
1M-K2-PROP	8	8	9	8.33333	0.55089
3M-K2-RANK	9	9	12	10	0.54014
3M-K2-EQ	11	11	8	10	0.52644
3M-K2-PROP	10	10	11	10.33333	0.53660
1M-K2-RANK	12	12	10	11.33333	0.50781

3M-K3-EQ ranks first, with the highest Sharpe (0.64198), the lowest volatility (0.09989) and the smallest tail loss (ES95 0.01574 per day). 1M-K3-EQ is a near tie (Sharpe 0.64014, best Calmar 0.20176); the two are 98% correlated, so the choice of tenor is immaterial. 3M-K3-EQ does *not* have the highest raw return: 3M-K2-RANK earns 0.07300 a year arithmetically, but with 35% more volatility and a -0.45106 drawdown. The difference in Sharpe between the top portfolios is small relative to its sampling error (roughly $1/\sqrt{11.4} \approx 0.3$ for an 11-year sample). So the ranking says which construction was more efficient in this sample, not that it will win in the future.

3.3 Discussion

Number of currencies is the dominant factor. Moving from K = 2 to K = 3 raises Sharpe in 6/6 pairs, lowers volatility by about 0.01946 a year and improves maximum drawdown by about 0.06787. The mean return barely changes. The 3rd-ranked currencies (GBP/NOK long, USD/SEK/EUR short) carry less than the extremes, but adding them diversifies the currency-specific crash risk of NZD, AUD and JPY. A K = 2 book is mostly a single NZD/AUD-vs-JPY/CHF bet. This is intuitive: carry's premium comes from a common factor, so extra names dilute idiosyncratic noise faster than they dilute the premium.

Allocation matters second. RANK and PROP tilt towards the most extreme carry (NZD and JPY), which raises the mean slightly (PROP has the highest CAGR within most groups) but raises volatility and tail risk more. EQ

has the best Sharpe for K = 3 at both tenors. Concentrating on the highest-carry pair concentrates exactly the exposure that crashes together in risk-off episodes.

Tenor matters least (average Sharpe range 0.01736, return correlations ≥ 0.980). 1M and 3M rates rank G-10 currencies almost identically; 3M rankings flip more often (Table 6) but produce nearly identical P&L. This is intuitive because cross-country rate differentials are persistent and similar across the short end of the curve.

Common to all 12: negative skewness (about –0.45 to –0.64), excess kurtosis of 7.5 to 11.6, worst days on 2008-10-24 or 2008-10-08, and a single drawdown episode (July 2007 to February 2009) that erases about four years of gains. This is the classic “up the stairs, down the elevator” carry profile. It motivates the volatility conditioning in parts (c) to (e).

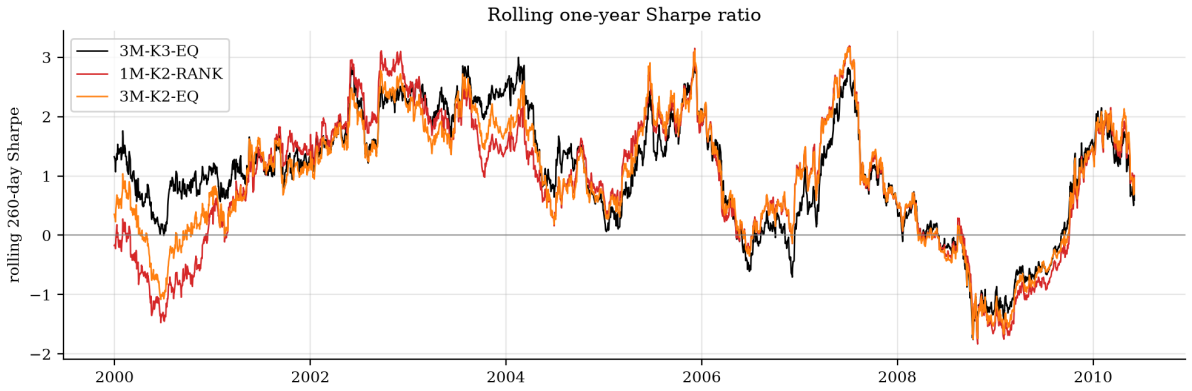


Figure 7: Rolling one-year Sharpe ratio of the best baseline versus two K = 2 portfolios.

4. Part (c): the best baseline (3M-K3-EQ) and volatility

I separate two questions. Contemporaneous association: does the carry return over (t, t+1] move with the *change* in volatility over the same interval? Predictive conditioning: does the volatility *level* known at t say anything about the carry return after t? Only the second can justify a trading rule. Volatility regressors are standardised (beta per one standard deviation), and t-statistics use Newey-West standard errors (5 lags daily, 21 lags for overlapping 21-day sums).

Table 13: Contemporaneous: R_{t+1} of 3M-K3-EQ vs the change in volatility from t to t+1.

vol series	corr(R, dVol)	corr(R, dlogVol)	beta (bp per 1 sd dVol)	HAC t	R2	n	share dVol = 0
AY1wk	-0.30901	-0.21003	-19.14583	-5.39870	0.09549	2979	0.53541
AY3M	-0.30141	-0.22465	-18.67456	-5.63609	0.09085	2979	0.18093
AY6M	-0.28551	-0.21192	-17.68960	-5.61581	0.08152	2979	0.20846
JPY3M	-0.37871	-0.34310	-23.46409	-8.75435	0.14342	2979	0.06311
AUD3M	-0.30249	-0.26680	-18.74172	-4.96745	0.09150	2979	0.09131
VIX	-0.49472	-0.39035	-31.03208	-13.50123	0.24474	2872	0.00522

Table 14: Predictive (linear): future returns regressed on the volatility level v_t (and its change dv_t). R 21d is the sum of the next 21 daily returns.

vol series	corr(v_t, R_{t+1})	beta 1d (bp/sd)	HAC t	corr(v_t, R 21d)	beta 21d (%/sd)	HAC t (21d)	corr(dv_t, R_{t+1})	HAC t
AY1wk	0.00015	0.00929	0.00384	-0.06375	-0.15956	-0.70962	0.11175	2.01518
AY3M	0.00071	0.04403	0.02204	-0.00266	-0.00667	-0.02812	0.08041	1.48064
AY6M	0.00659	0.40816	0.22434	0.02658	0.06662	0.29072	0.09884	2.02280
JPY3M	0.02000	1.23892	0.72697	0.04570	0.11389	0.57486	0.06911	1.50848
AUD3M	-0.00698	-0.43230	-0.20514	-0.03938	-0.09843	-0.41663	0.05952	1.31969
VIX	0.01169	0.72434	0.36047	0.02052	0.05124	0.22270	0.07144	1.81155

Table 15: Regimes: next-day carry statistics by full-sample tercile of the volatility level at t. Terciles use the whole sample, so this is a descriptive in-sample split, not a tradable rule.

vol / tercile of v_t	n	v min	v max	ann. mean R_{t+1}	ann. vol	Sharpe	ES95/day	worst day
AY1wk low	994	5.25000	10.60000	0.08039	0.06276	1.28101	0.00918	-0.02083
AY1wk medium	1192	10.63750	16	0.07367	0.07408	0.99444	0.01123	-0.02413
AY1wk high	793	16.04500	89.26750	0.03024	0.15599	0.19386	0.02481	-0.05182
AY3M low	1008	6.20000	10.30000	0.09593	0.06640	1.44474	0.01008	-0.02083
AY3M medium	978	10.32500	15.02000	0.05705	0.06868	0.83063	0.01019	-0.01703
AY3M high	993	15.02500	51.87750	0.03949	0.14434	0.27359	0.02305	-0.05182
AY6M low	999	6.75000	10.40000	0.08466	0.06760	1.25227	0.01040	-0.02083
AY6M medium	992	10.41190	15	0.05794	0.07276	0.79636	0.01102	-0.03153
AY6M high	988	15.07500	41.91000	0.05026	0.14204	0.35381	0.02262	-0.05182
JPY3M low	1007	6.10040	9.30000	0.08430	0.07011	1.20250	0.01108	-0.02083
JPY3M medium	986	9.30040	11.50000	0.03739	0.07869	0.47509	0.01231	-0.03153
JPY3M high	986	11.50500	27.38750	0.07094	0.13769	0.51523	0.02170	-0.05182
AUD3M low	993	6.35000	10	0.10857	0.06470	1.67797	0.00952	-0.02083
AUD3M medium	995	10.01500	12.15000	0.05531	0.07556	0.73202	0.01144	-0.03153
AUD3M high	991	12.18500	33.93500	0.02912	0.14171	0.20548	0.02266	-0.05182
VIX low	994	9.89000	17.77000	0.07902	0.06664	1.18571	0.01010	-0.02067
VIX medium	993	17.79000	24.24000	0.01640	0.07817	0.20984	0.01210	-0.02413
VIX high	992	24.25000	80.86000	0.09766	0.13933	0.70091	0.02265	-0.05182

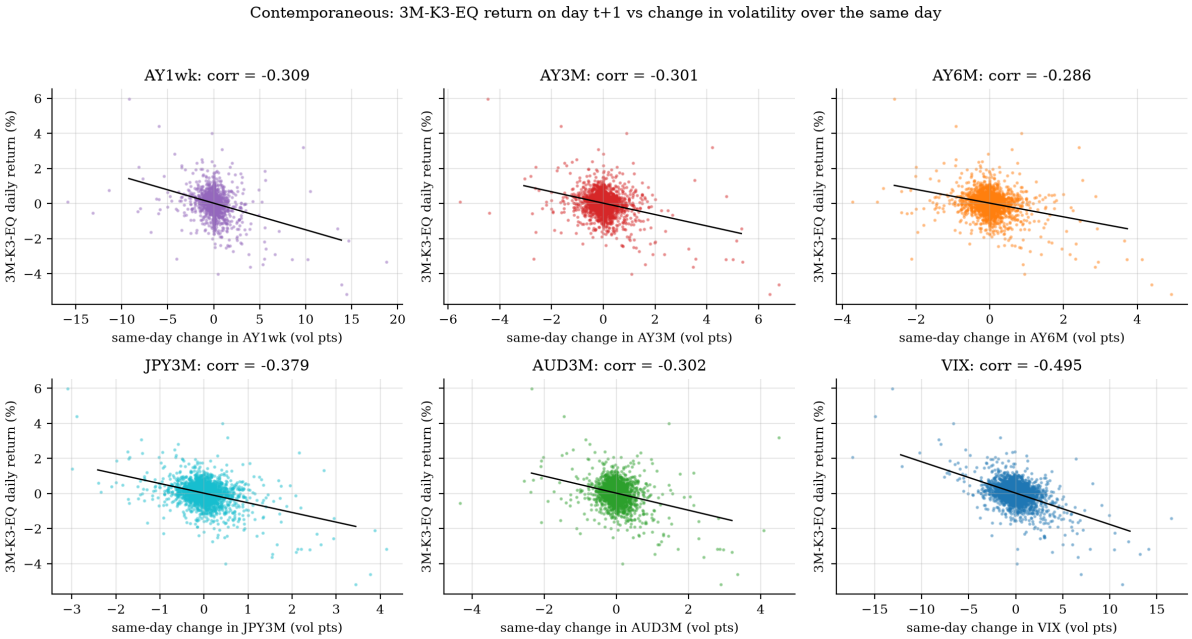


Figure 8: Contemporaneous scatter: daily carry return vs the same-day change in each volatility series, with OLS fit.

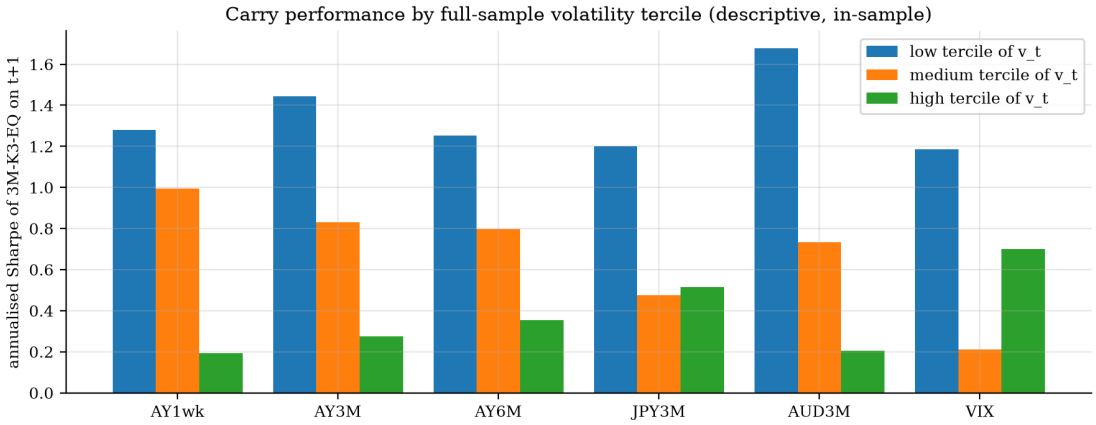


Figure 9: Next-day Sharpe ratio by volatility tercile.

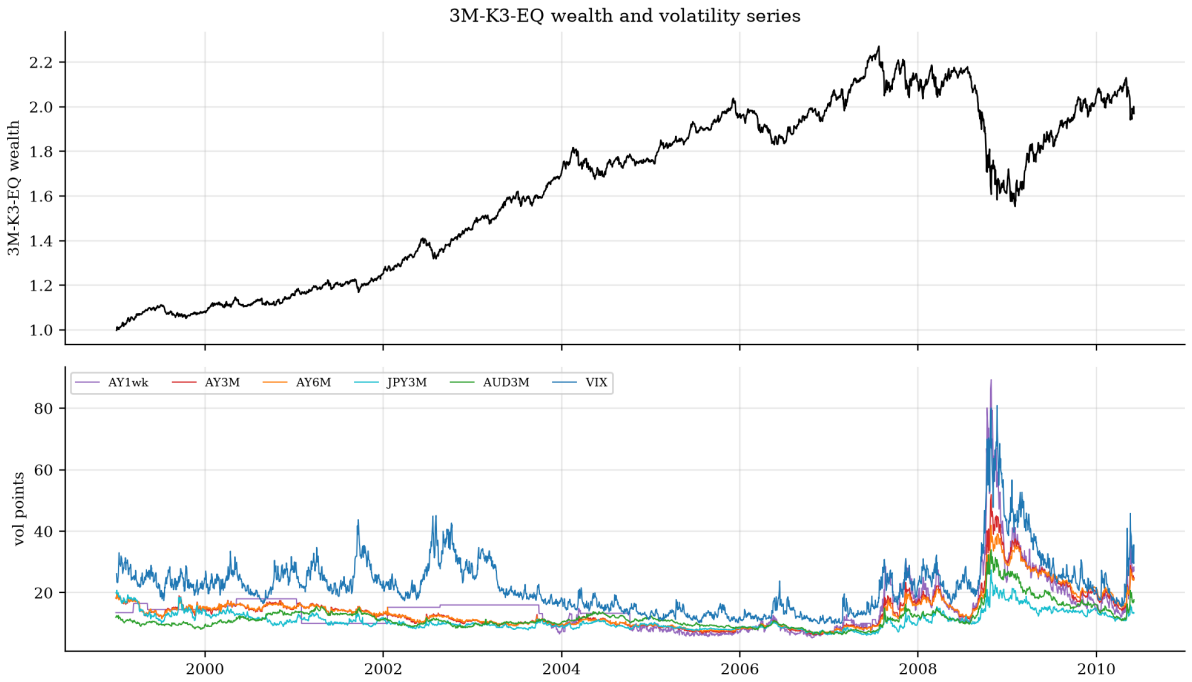


Figure 10: Best-baseline wealth and the six volatility series.

Table 16: Candidate ranking. Primary criterion = low-minus-high tercile Sharpe spread of R_{t+1} (predictive information only); tie-break = smaller high-regime ES. Other columns are shown for context.

vol series	Sharpe low - high	Sharpe low	Sharpe high	ES95 high regime	HAC t (21d level)	contemp. corr
AUD3M	1.47249	1.67797	0.20548	0.02266	-0.41663	-0.30249
AY3M	1.17114	1.44474	0.27359	0.02305	-0.02812	-0.30141
AY1wk	1.08715	1.28101	0.19386	0.02481	-0.70962	-0.30901
AY6M	0.89846	1.25227	0.35381	0.02262	0.29072	-0.28551
JPY3M	0.68727	1.20250	0.51523	0.02170	0.57486	-0.37871
VIX	0.48480	1.18571	0.70091	0.02265	0.22270	-0.49472

4.1 Discussion

Contemporaneously, carry is clearly short volatility. All six correlations are negative and highly significant (HAC $|t| \geq 4.97$). The VIX change has the strongest link (corr -0.49472, R^2 0.24474), followed by JPY3M (-0.37871). Days when equity and yen volatility jump are the days when funding currencies rally and investment currencies fall. The AUD/JPY 1-week series has a zero change on 54% of days (stale quotes), which weakens its measured association.

Linear predictability is absent. The level of volatility at t has essentially zero correlation with R_{t+1} ($|\text{corr}| \leq 0.020$) and no significant 21-day slope ($|t| \leq 0.71$). Mean returns are not linearly lower when volatility is high. The 2008 crash is followed by large 2009 rebounds while volatility was still very high, and these offset each other in a linear fit.

The regime split shows what matters for conditioning. High volatility mainly means much higher *risk* (annualised volatility of R_{t+1} roughly doubles, ES95 more than doubles) with a lower mean. The Sharpe drop from the low to the high tercile is largest for AUD3M (1.67797 to 0.20548), then AY3M (1.44474 to 0.27359) and AY1wk. VIX and JPY3M are non-monotonic: their high tercile has a *higher* Sharpe than the medium one (VIX 0.70091 vs 0.20984), because extreme equity or yen volatility often marks the rebound phase after a crash.

Candidate: AUD3M, the 3-month AUD/USD implied vol. It is intuitive. AUD is held long on 75% of days and is the archetypal investment currency, so option-implied uncertainty about AUD measures the risk of the portfolio's largest long exposure directly. The AUD/JPY vols (AY3M in particular) are a close second; they price the canonical carry cross, long the top investment currency against the top funding currency. A 3-month tenor is a compromise between responsiveness and noise. The 1-week tenor reacts fastest but is stale before 2005, and the 6-month tenor is the smoothest and slowest. Equity vol (VIX) reflects the broad risk-off driver of carry unwinds, but it also spikes for equity-specific reasons (2001–2003) when carry did well.

4.2 Relationship to Jurek

Limitation: the Jurek reading was not in the materials available to me, so I cannot verify its specific statements or numbers, and I attribute none to it. I test the relationship as the assignment frames it: FX carry returns versus FX implied volatility and equity (VIX) volatility. My data confirm a strong contemporaneous link: carry losses coincide with jumps in FX implied vol and in the VIX, and the VIX co-moves most strongly. This fits the view that carry is exposed to global risk-off and crash risk. For my portfolio the predictive link is weaker and comes through risk rather than mean: high-volatility states are not reliably followed by lower average returns in a linear sense, but they are followed by far worse risk-adjusted returns and fatter left tails. On that dimension FX vol tied to the portfolio's own currencies beats equity vol.

5. Part (d): conditioning on each implied-volatility series

The same rule is applied to each of the six variables, with no optimisation:

$$q_t = Q_{0.75}\big(\{v_s : \ s < \tau(t)\}\big), \qquad s_t = \mathbf{1}\{v_{\tau(t)} \le q_t\}, \qquad R^c_{t+1} = s_t\, R_{t+1}$$

Here τ(t) is the latest date ≤ t on which the variable was observed. The threshold is the expanding 75th percentile of all *earlier* observations: the current observation enters the threshold only from the next day on, i.e. it is added after the signal is computed. At least 252 observations (about one year) are required before the first signal. When $s_t = 0$ the carry book is flat and the excess return is exactly zero. The cut-off is the same round number for every variable, and the 75th percentile simply means 'volatility is in the top quarter of its own history so far'. Because all six series are compared on the same dates, they and the baseline share a common sample from 2000-01-04 to 2010-06-04 (2719 days), which starts after the VIX warm-up.

Table 17: 3M-K3-EQ unconditioned vs conditioned on each implied-volatility series, 2000-01-04 to 2010-06-04.

strategy	ann. mean	CAGR	ann. vol	Sharpe	skew	ex. kurt	max DD	DD trough	ES95/day	worst day	Calmar
baseline	0.06298	0.05938	0.10282	0.61256	-0.49154	10.94975	-0.31637	2009-02-02	0.01629	-0.05182	0.18768
AY1wk-cond	0.06056	0.06056	0.05924	1.02224	-0.75321	3.99746	-0.14722	2010-02-23	0.00944	-0.02413	0.41135
AY3M-cond	**0.06919**	**0.06978**	**0.05873**	**1.17806**	**-0.63027**	**3.49345**	**-0.10212**	**2006-05-22**	**0.00921**	**-0.02083**	**0.68329**
AY6M-cond	0.06189	0.06197	0.05936	1.04264	-0.63768	3.34183	-0.11208	2010-04-16	0.00937	-0.02083	0.55289
JPY3M-cond	0.04867	0.04753	0.06686	0.72805	-1.02633	5.88322	-0.23228	2010-03-04	0.01084	-0.03180	0.20460
AUD3M-cond	0.05427	0.05407	0.05676	0.95624	-1.06132	9.08738	-0.10212	2006-05-22	0.00914	-0.03180	0.52944
VIX-cond	0.05026	0.04931	0.06501	0.77299	-0.87097	5.42174	-0.14659	2008-09-11	0.01059	-0.03180	0.33639

Table 18: Exposure behaviour. A 'switch' is a change between invested and flat. % days > 0 counts flat days as non-positive.

strategy	% days invested	switches	off spells	avg off-spell (days)	% days > 0	% invested days > 0	DD peak	DD recovery
baseline	100	0	0	0.00	55.39	55.39	2007-07-25	not recovered
AY1wk-cond	73.34	37	19	38.16	41.01	55.92	2007-07-25	not recovered
AY3M-cond	**74.70**	**57**	**29**	**23.72**	**41.96**	**56.18**	**2005-12-05**	**2007-01-11**
AY6M-cond	75.40	55	28	23.89	42.04	55.76	2007-07-25	not recovered
JPY3M-cond	81.54	43	22	22.82	45.20	55.44	2007-07-25	not recovered
AUD3M-cond	63.30	93	47	21.23	35.53	56.13	2005-12-05	2007-01-11
VIX-cond	75.03	115	58	11.71	41.63	55.49	2007-07-25	not recovered

Table 19: Same decision rule as in part (b), applied to the six conditioned strategies.

strategy	Sharpe rank	Calmar rank	ES95 rank	average rank	Sharpe
AY3M-cond	**1**	**1**	**2**	**1.33333**	**1.17806**
AY6M-cond	2	2	3	2.33333	1.04264
AUD3M-cond	4	3	1	2.66667	0.95624
AY1wk-cond	3	4	4	3.66667	1.02224
VIX-cond	5	5	5	5	0.77299
JPY3M-cond	6	6	6	6	0.72805

Table 20: Stress episodes: cumulative return within each window and the first date the signal switched carry off (searching from 45 calendar days before the window).

strategy	Aug-2007 return	Sep-Dec 2008 return	Apr-Jun 2010 return	first off (Aug-07)	first off (Sep-08)	first off (Apr-10)
baseline	-0.06657	-0.20121	-0.05669			
AY1wk-cond	-0.03622	-0.00830	0.01149	2007-08-09	2008-08-13	2010-03-01
AY3M-cond	-0.04495	0.00000	0.01501	2007-08-14	2008-08-13	2010-03-01
AY6M-cond	-0.05589	-0.00830	0.00398	2007-08-15	2008-08-26	2010-03-01
JPY3M-cond	-0.07571	-0.05400	-0.03358	2007-08-16	2008-07-18	2010-05-06
AUD3M-cond	-0.07842	0.00000	-0.02068	2007-08-16	2008-08-11	2010-03-01
VIX-cond	-0.04421	-0.03696	-0.01728	2007-08-03	2008-09-09	2010-05-06

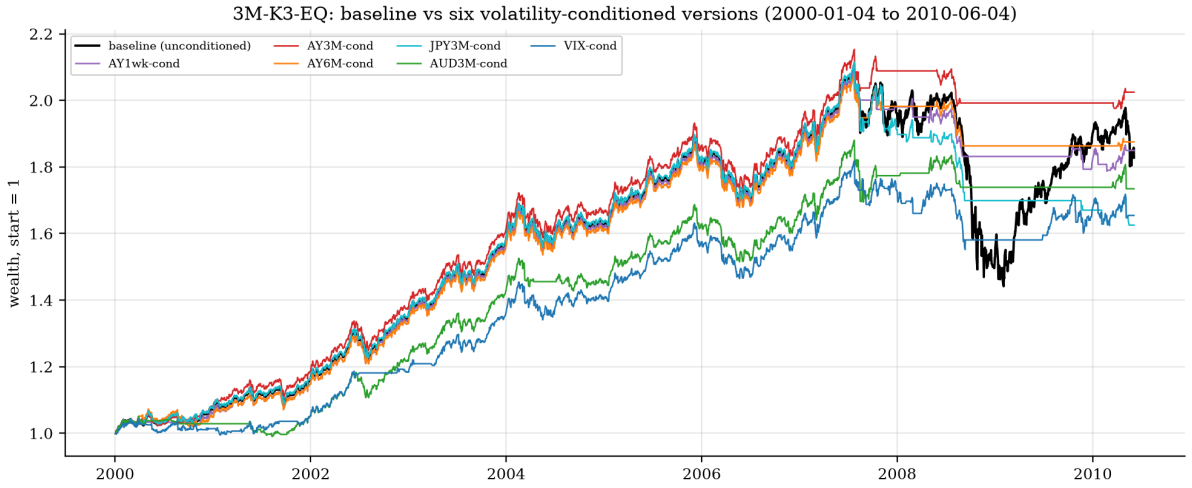


Figure 11: Wealth of the baseline and the six conditioned strategies on the common sample.

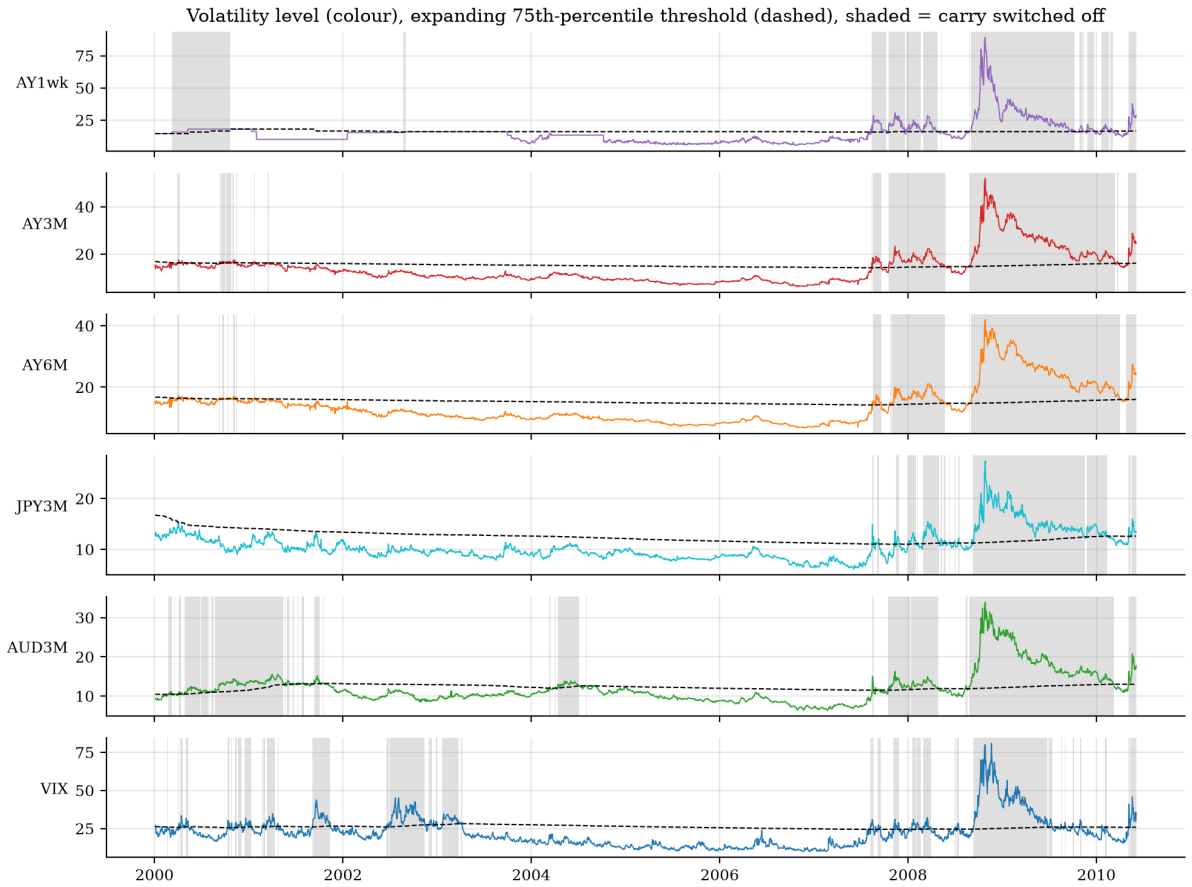


Figure 12: Each volatility series, its real-time expanding 75th-percentile threshold, and the periods with carry switched off (shaded).

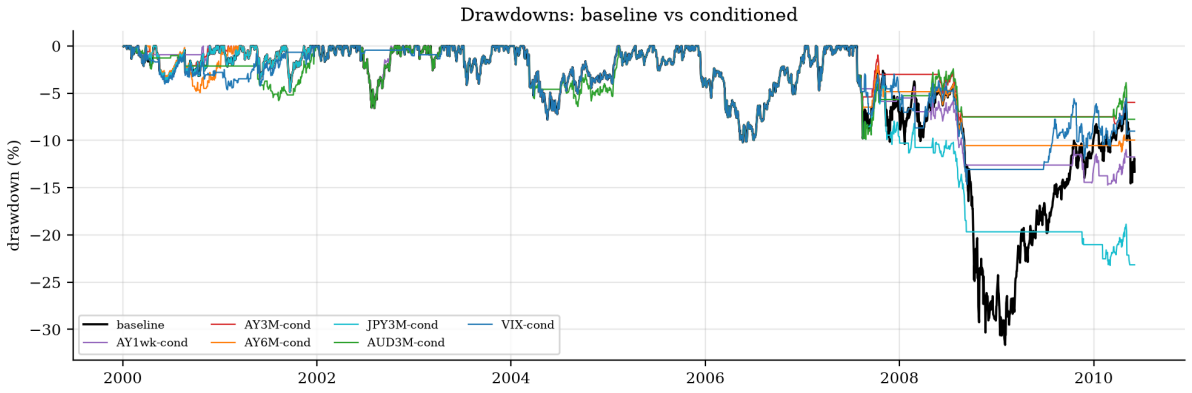


Figure 13: Drawdowns of the baseline and conditioned strategies.

5.1 Discussion

Best: AY3M-cond (the 3-month AUD/JPY implied vol). Sharpe 1.17806 vs 0.61256 for the baseline, CAGR 0.06978 vs 0.05938, volatility 0.05873 vs 0.10282, maximum drawdown -0.10212 vs -0.31637, excess kurtosis 3.49345 vs 10.94975, invested 74.70% of days. It was switched off from 2008-08-13, before Lehman, and sat out the whole autumn 2008 crash, so its maximum drawdown is a 2006 episode instead. The ranking is AY3M, AY6M, AUD3M, AY1wk, VIX, JPY3M. Every conditioned strategy beats the baseline on Sharpe, volatility and drawdown.

(i) Agreement with (c). Partial. Part (c) singled out AUD3M with AY3M a close second, and the AUD-linked FX implied vols (AY3M, AY6M, AUD3M, AY1wk) do occupy the top four places. VIX and JPY3M, the two variables with the strongest contemporaneous correlation, rank last, as the non-monotonic regime results predicted. But AUD3M itself is third (Sharpe 0.95624). It has the lowest ES and the same drawdown as AY3M-cond, but it switches 93 times and is invested only 63.30% of the time. Its real-time threshold flagged 2000–2001 and 2004 as high-vol periods when carry earned well (Figure 12). It also turned off only after the worst August 2007 losses and partly missed the late-August rebound (-0.07842 in that episode, worse than the baseline).

(ii) Agreement with intuition. Yes, broadly. The 3-month AUD/JPY vol prices the exact risk of the carry book's core exposure (long AUD/NZD against JPY), at a tenor that is responsive but not noisy. Equity vol is a less specific signal: it was elevated in 2001–2003 (Figure 12, VIX switched off for long stretches) for equity-specific reasons while carry did well, and it has the most switches (115). JPY3M (USD/JPY vol) ranks last. Yen volatility mixes in USD/JPY-specific moves, and although it first switched off early (July 2008) it whipsawed back in during September 2008 and lost -0.05400 in Sep–Dec 2008. The 1-week tenor carries little information before 2005 because its quotes are stale.

(iii) Why test all six. Part (c) is descriptive and in-sample: it uses full-sample terciles, linear correlations and contemporaneous changes. A trading rule depends on things (c) does not measure: *when* the real-time threshold is crossed, how long the variable stays above it, how often it whipsaws, whether the off periods also miss the rebounds, and data quality such as staleness. Contemporaneous correlation is not predictability: VIX has the strongest association yet the second-worst rule. The ranking of conditioned strategies is therefore an empirical question that must be checked variable by variable. Picking the best of six is itself selection, so I also checked robustness (§7, Table 26): AY3M stays first with an extra day of signal lag and (tied with AY1wk) with a 90th-percentile threshold, and the AY3M rule improves all 12 baselines, not just the one it was selected for.

6. Part (e): conditioning on historical (realised) volatility

For the historical measure I use the 21-observation realised volatility of AUD/JPY, the canonical carry cross (long the archetypal investment currency, short the archetypal funding currency). The best baseline is long AUD on most days and short JPY on every day. AUD/JPY is also the pair underlying the AY implied vols, so this is a direct realised-versus-implied comparison on the same asset. 21 trading days (about one month) roughly matches the 30-day VIX horizon and the 1M rate tenor. The cross is built from the supplied spot data on standardised levels, and only past and current returns enter the estimate:

$$P_t = \frac{X_{AUD,t}}{X_{JPY,t}} \ (\text{JPY per AUD}), \qquad RV_t = 100\sqrt{260}\ \text{sd}\Big(\ln \frac{P_s}{P_{s-1}},\ s = t-20, \ldots, t\Big)$$

Log returns are used only to estimate volatility; strategy returns stay arithmetic. The conditioning rule is exactly the part (d) rule (expanding 75th percentile of prior RV values, 252-observation warm-up, flat when above). The comparison sample from 2000-01-20 to 2010-06-04 is common to all three strategies and starts after the RV warm-up.

Table 21: Baseline vs best implied-vol rule (AY3M-cond) vs realised-vol rule, 2000-01-20 to 2010-06-04.

strategy	ann. mean	CAGR	ann. vol	Sharpe	ex. kurt	max DD	DD trough	ES95/day	Calmar	% invested	switches
baseline	0.06125	0.05753	0.10300	0.59469	10.91013	-0.31637	2009-02-02	0.01629	0.18183	100	0
AY3M-cond	0.06749	0.06796	0.05878	1.14809	3.49087	-0.10212	2006-05-22	0.00921	0.66547	74.58441	57
RV21_AUDJPY-cond	0.05419	0.05378	0.05989	0.90474	4.51258	-0.13562	2009-08-14	0.00956	0.39657	74.43665	59

Table 22: Stress episodes for the part (e) comparison.

strategy	Aug-2007 return	Sep-Dec 2008 return	Apr-Jun 2010 return	first off (Aug-07)	first off (Sep-08)	first off (Apr-10)
baseline	-0.06657	-0.20121	-0.05669			
AY3M-cond	-0.04495	0.00000	0.01501	2007-08-14	2008-08-13	2010-03-01
RV21_AUDJPY-cond	-0.02827	-0.05034	0.01501	2007-07-27	2008-09-09	2010-03-01

Table 23: Lead-lag between daily changes in RV and in AY3M. Positive L = implied vol change L days earlier. Correlations at L > 0 exceed those at L < 0: implied vol tends to move first.

lag L (days)	-5	-4	-3	-2	-1	0	1	2	3	4	5
corr(dRV$_t$, dAY3M$_{t-L}$)	0.005	0.028	0.009	0.052	0.055	0.186	0.136	0.116	0.133	0.119	0.092

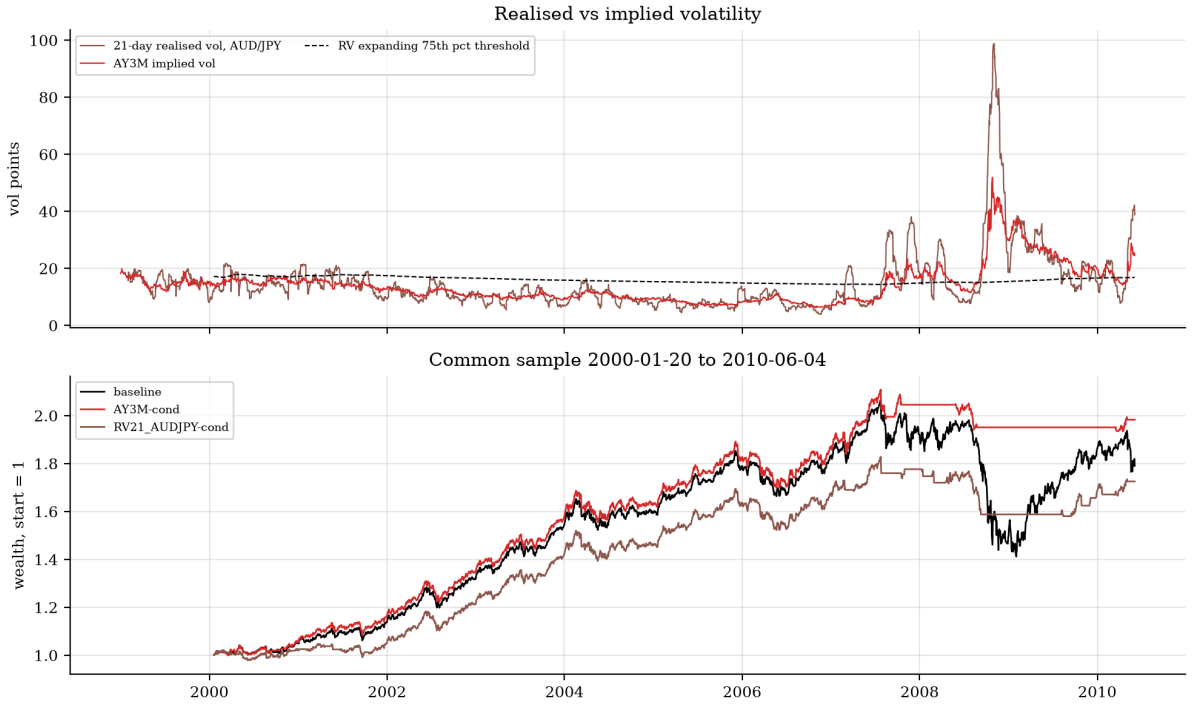


Figure 14: Top: AUD/JPY 21-day realised vol, AY3M implied vol and the RV threshold. Bottom: wealth on the common sample.



Figure 15: Cumulative returns within the three stress episodes.

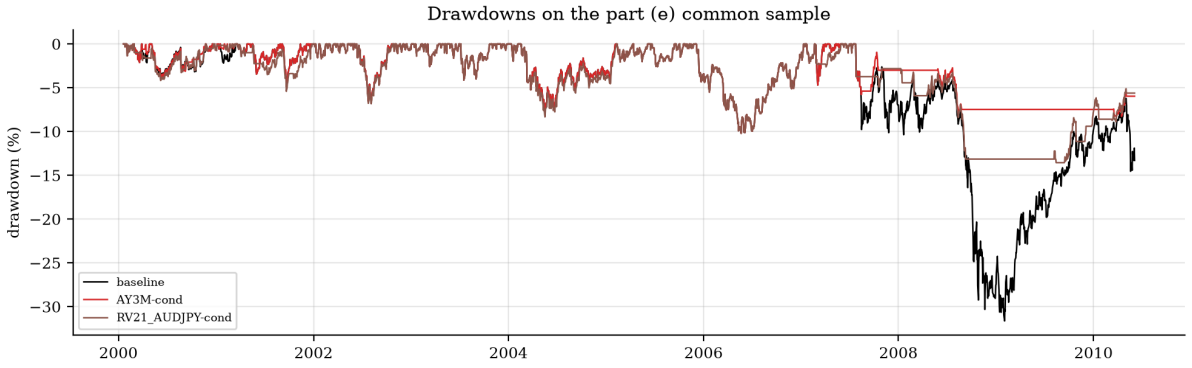


Figure 16: Drawdowns on the part (e) common sample.

6.1 Discussion

Realised-vol conditioning helps, but less than implied vol. Versus the baseline, RV conditioning raises Sharpe from 0.59469 to 0.90474, cuts volatility from 0.10300 to 0.05989 and maximum drawdown from -0.31637 to -0.13562. AY3M-cond does better on every headline metric (Sharpe 1.14809, CAGR 0.06796 vs 0.05378, maximum drawdown -0.10212, ES95 0.00921 vs 0.00956). Both are invested about 74.5% of the time, and their on/off states agree on 86.7% of days, so the difference comes from *timing*, not from how often each is invested.

Backward-looking vs forward-looking. RV is a backward-looking average of the last month's squared moves, so it rises only after large moves have occurred. Implied vol is a forward-looking market price that also includes option buyers' demand for crash protection (a risk premium), so it can rise before realised moves do. Table 23 supports this: implied-vol changes lead RV changes. In 2008 the AY3M signal switched off on 2008-08-13, while RV crossed its threshold on 2008-09-09, so the RV rule lost -0.05034 in Sep–Dec 2008 versus 0.00000. RV also falls back quickly once daily moves calm down. The RV rule re-entered carry in August 2009, while the AY3M rule stayed flat until March 2010 because option markets kept pricing elevated risk. The RV rule's maximum drawdown runs from 2007-07-25 to 2009-08-14 and includes the September 2008 losses taken before it switched off.

But not uniformly. In the sudden August 2007 unwind the RV signal switched off earlier (2007-07-27 vs 2007-08-14) and lost less (-0.02827 vs -0.04495). Before 2007 AUD/JPY implied vol was low and sticky (partly stale), whereas a few large daily moves immediately lift a 21-day RV. Responsiveness to stress therefore depends on the episode. RV reacts fast to realised shocks; implied vol can anticipate a build-up of risk. Over the whole sample the forward-looking measure produced better risk-adjusted results, which agrees with my intuition, but the evidence comes from a handful of episodes and should not be read as a guarantee.

7. Verification, reconciliation and sensitivity

7.1 Manual reconciliation of one interval

Table 24: 3M-K3-EQ from 2008-10-24 (Friday close) to 2008-10-27 (Monday): weights from 3M carry at 2008-10-24; $\Delta = 3/360$; $e = (1+r)(1+i\Delta) - (1+i_{USD}\Delta)$; contribution = w × e. Sum of contributions = -0.0230153417, identical to the pipeline value -0.0230153417.

item	JPY	CAD	AUD	NZD	CHF	NOK
quote_convention	JPY per USD	CAD per USD	USD per AUD	USD per NZD	CHF per USD	NOK per USD
quote_t	94.320000	1.277500	0.622500	0.556500	1.167400	6.995100
quote_t+1	92.780000	1.295300	0.601300	0.541400	1.155400	6.965700
X_t (USD/ccy)	0.010602	0.782779	0.622500	0.556500	0.856604	0.142957
X_t+1 (USD/ccy)	0.010778	0.772022	0.601300	0.541400	0.865501	0.143561
spot_return	0.016598	-0.013742	-0.034056	-0.027134	0.010386	0.004221
rate_t_pct	0.940000	3.062600	6.250000	7.750000	2.340000	6.725000
usd_rate_t_pct	4.110000	4.110000	4.110000	4.110000	4.110000	4.110000
delta	0.008333	0.008333	0.008333	0.008333	0.008333	0.008333
excess_return	0.016336	-0.013833	-0.033896	-0.026848	0.010241	0.004441
weight	-0.333333	-0.333333	0.333333	0.333333	-0.333333	0.333333
contribution	-0.005445	0.004611	-0.011299	-0.008949	-0.003414	0.001480

Check, e.g. JPY: the quote falls from 94.32 to 92.78 JPY per USD, so X_{JPY} rises by 94.32/92.78 − 1 = 0.016598 (the yen appreciates). The accrual adds $(1.016598)(1 + 0.0094\times3/360) - (1 + 0.0411\times3/360)$, giving e = 0.016336. Weighted by −1/3, this costs the short-yen book 0.005445 that day.

7.2 Automated checks

– Dates strictly increasing, no duplicates, weekdays only; worksheet headers match the assignment's column letters.

– Quote inversion: X_{JPY} = 1/94.32 on 2008-10-24; computed spot returns equal the workbook's L–T columns (max error 2.2e-16); negating raw-quote returns is shown to be wrong. The AUD/JPY cross equals the AUD quote × the JPY quote.

– Rates are decimals; Friday→Monday $\Delta = 3/360$ and weekday $\Delta = 1/360$; the excess-return formula is reproduced by hand.

– Weights for all 12 strategies on every day: K longs and K shorts, +1/–1 sums, longs = top-K carry, shorts = bottom-K; PROP magnitudes stay non-negative when all carries share a sign; the zero-denominator fallback works; RANK = (3,2,1)/6.

– No look-ahead: changing all rates and spot data after a cut-off date leaves earlier weights and returns unchanged; changing volatility after a cut-off leaves earlier thresholds and signals unchanged; the threshold at date d equals the 75th percentile of observations strictly before d; the first 251 thresholds are missing (warm-up).

– Timing: the conditioned return at t+1 uses the signal at t; the first conditioned return is missing, not zero.

– Compounding, geometric mean, annualisation and drawdown (peak, trough, recovery) checked on a hand-computed series.

– No artificial zero returns: the 12 baselines have 0 exactly-zero days and no gaps. VIX is not filled before 1999-01-04. RV uses only the current and previous 20 returns.

– Common samples: part (b) uses one sample for all 12; part (d) uses one sample for the baseline and six rules; part (e) uses one sample for all three strategies.

7.3 Sensitivity to conventions

Table 25: USD as a constituent vs pure funding currency (9-currency ranking). The best baseline is unchanged (3M-K3-EQ), and K = 3 still dominates.

portfolio	Sharpe (USD in universe)	Sharpe (USD excluded)	max DD (USD in)	max DD (USD excluded)
1M-K2-EQ	0.55609	0.54823	-0.38264	-0.37959
1M-K2-RANK	0.50781	0.50204	-0.43032	-0.42846
1M-K2-PROP	0.55089	0.54348	-0.41021	-0.40709
1M-K3-EQ	0.64014	0.64118	-0.31171	-0.29614
1M-K3-RANK	0.58139	0.57489	-0.37220	-0.36382
1M-K3-PROP	0.60458	0.58875	-0.36566	-0.35825
3M-K2-EQ	0.52644	0.51810	-0.42071	-0.40262
3M-K2-RANK	0.54014	0.50586	-0.45106	-0.43972
3M-K2-PROP	0.53660	0.52465	-0.44168	-0.42933
3M-K3-EQ	0.64198	0.64879	-0.31637	-0.29502
3M-K3-RANK	0.59689	0.57877	-0.38565	-0.36947
3M-K3-PROP	0.61510	0.59537	-0.37781	-0.36337

Switching the day count from ACT/360 to ACT/365 for all currencies moves the 3M-K3-EQ Sharpe from 0.64198 to 0.63601 and the CAGR from 0.06091 to 0.06027. This is immaterial.

Table 26: Robustness of the conditioning conclusions (not used to choose the rule). The extra lag guards against the FX and VIX closes not being synchronous. AY3M remains first in each column (tied with AY1wk at q90), and AUD/JPY-related FX vols stay ahead of VIX and JPY3M. Implied-vol rows and the baseline row use the part (d) sample; the RV21_AUDJPY row uses the part (e) sample (2000-01-20 onwards).

signal variable	Sharpe: q75, no extra lag (primary)	Sharpe: q75, signal lagged 1 more day	Sharpe: q90, no extra lag
AY1wk	1.02323	0.91550	1.00099
AY3M	1.17909	1.01559	1.00126
AY6M	1.04364	0.98476	0.87556
JPY3M	0.72890	0.75832	0.72357
AUD3M	0.95726	0.81415	0.89050
VIX	0.77387	0.77325	0.82192
RV21_AUDJPY	0.90406	0.87842	0.86303
baseline	0.61313	0.61313	0.61313

Table 27: The AY3M rule applied to every baseline over 2000-01-04 to 2010-06-04. It improves all 12, so the result is not specific to the portfolio it was chosen for.

baseline	Sharpe unconditioned	Sharpe AY3M-conditioned	max DD unconditioned	max DD AY3M-conditioned
1M-K2-EQ	0.57543	1.04591	-0.38264	-0.11802
1M-K2-RANK	0.55486	1.06605	-0.43032	-0.14120
1M-K2-PROP	0.57709	1.06488	-0.41021	-0.13676
1M-K3-EQ	0.60888	1.17680	-0.31171	-0.10624
1M-K3-RANK	0.59303	1.15823	-0.37220	-0.12164
1M-K3-PROP	0.60425	1.18335	-0.36566	-0.12408
3M-K2-EQ	0.54536	1.08559	-0.42071	-0.11831
3M-K2-RANK	0.59323	1.14745	-0.45106	-0.14141
3M-K2-PROP	0.56753	1.11712	-0.44168	-0.13788
3M-K3-EQ	0.61256	1.17806	-0.31637	-0.10212
3M-K3-RANK	0.61438	1.20772	-0.38565	-0.11971
3M-K3-PROP	0.62102	1.21670	-0.37781	-0.12232

8. Assumptions and limitations

– Course definitions not available. No lecture notes were in the workspace, so the definitions of rank-based and proportional-to-carry weights (§1.5), the +1/–1 exposure normalisation, ACT/360 accrual and the 260-day annualisation are my explicit assumptions, not conventions specified by the professor.

– USD treated as a ranking constituent (§1.4); excluding it does not change conclusions (Table 25).

– Rate tenor as an overnight rate. I accrue the 1M or 3M annualised rate daily, as if the deposit could be rolled daily at that rate. A real 3M deposit locks its rate for three months, and forward points would differ slightly.

– Excess returns only. No collateral interest is added and no transaction costs are charged (as instructed). The conditioned strategies switch 37–115 times, and the baselines trade whenever rankings or PROP weights change, so costs would reduce all results, the conditioned ones somewhat more.

– Timing. The FX, rate and volatility data are treated as known at the same daily close. The sources' exact fixing times are not documented in the workbook (VIX is a US close). An extra day of lag (Table 26) leaves the conclusions unchanged.

– Data quality. AY1wk is stale for most of 1999–2004 and AY3M/AY6M are partly stale before 2008. I used the data as supplied and did not clean or replace any series.

– In-sample evidence. One 11.4-year sample dominated by a single crash (2008). Choosing the best of 12 baselines and 6 rules is in-sample selection; Sharpe differences among top candidates are well within sampling error. Nothing here implies out-of-sample superiority.

– Jurek reading unavailable; no specific claims are attributed to it (§4.2).

9. Reproduction

From the hw3 directory (Python 3.12; package versions pinned in requirements.txt):

```
python3.12 -m venv .venv && .venv/bin/pip install -r requirements.txt

.venv/bin/python scripts/run_all.py   # analysis: tables, histories, figures, results.json

.venv/bin/python -m pytest -q tests   # 16 verification tests

.venv/bin/python scripts/build_report.py   # this PDF
```

Code: src/fxcarry/data.py (loading, layout check, quote standardisation), returns.py (spot and excess returns), portfolios.py (ranking, weights, validation), metrics.py (statistics, drawdowns, decision rule), volanalysis.py (part c), conditioning.py (thresholds, signals, realised vol). Outputs: outputs/tables (all tables as CSV plus hw3_tables.xlsx), outputs/histories (daily weights, returns, signals), outputs/figures.